Ex 6b: Time Series Data Analysis and Forecast

Question

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Department of Artificial Intelligence and Data Science
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AD23532 – Principles of Data Science

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Ex6b: Time Series Data Analysis and Forecast

Learning Objective:
The objective of this experiment is to enable students to understand and implement the fundamental concepts of time series data analysis and forecasting using Python.

Problem Statement:
An airline company maintains monthly records of the number of passengers travelling by air. The company wants to understand historical passenger trends and seasonal patterns and use this information to forecast future passenger demand.

As a Data Scientist, analyse the historical monthly airline passenger data, determine whether the time series is stationary, apply appropriate transformations when necessary, build an ARIMA forecasting model, predict passenger numbers for the test period, and evaluate the accuracy of the forecasts.

Tasks to Perform:

1. Download airline-passenges dataset from https://github.com/jbrownlee/Datasets/blob/master/airline-passengers.csv
2. Read the CSV with parse_dates and index_col so the time index becomes a DatetimeIndex.
3. Explore the Dataset.
4. Plot the series to inspect trend/seasonality/outliers.
5. Do Stationarity check by running Augmented Dickey-Fuller (ADF) test.
6. If the series is non-stationary, convert it to stationary by differencing the series.
7. Re-run ADF on the differenced series and visualize the new dataframe.
8. If still non-stationary, consider another difference (d=2) or transformation (log) before differencing.
9. Split the dataset for training and testing on original series.
10. Fit ARIMA model on training set. d value comes from Step 8 (how many differences you applied).
11. Perform forecasts on the test dataset.
12. Compute metrics MSE, MAE, RMSE on the predictions.
13. Plot train, test, and predictions in a single plot.

Aim

To analyse the monthly airline passenger time series, test it for stationarity, make it stationary by differencing, and build and evaluate an ARIMA model for forecasting passenger numbers.

Procedure

1. Download airline-passengers.csv and read it with parse_dates and index_col so the index is a DatetimeIndex.
2. Explore the dataset and plot the series to inspect trend, seasonality and outliers.
3. Run the Augmented Dickey-Fuller (ADF) test; if the p-value is above 0.05 the series is non-stationary.
4. Apply differencing (and a log transform / second difference if required) and re-run the ADF test; plot the differenced series.
5. Split the original series into training and testing sets.
6. Fit an ARIMA model on the training set using the d value found above and forecast the test period.
7. Compute MSE, MAE and RMSE, and plot train, test and predictions together.

Code

```
#Mohammed Fayyad(241801165)
import pandas as pd, numpy as np, matplotlib.pyplot as plt
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.arima.model import ARIMA
from sklearn.metrics import mean_squared_error, mean_absolute_error

df = pd.read_csv("airline-passengers.csv", parse_dates=["Month"], index_col="Month")
print(df.head()); print(df.shape); df.info(); print(df.describe())
df.plot(figsize=(10, 4)); plt.title("Monthly Airline Passengers"); plt.show()

def adf(series, label=""):
    r = adfuller(series.dropna())
    print(f"{label} ADF stat={r[0]:.4f}  p-value={r[1]:.4f}")
    print("Stationary" if r[1] < 0.05 else "Non-stationary")

series = df["Passengers"]
adf(series, "Original")

d1 = series.diff().dropna();  adf(d1, "d=1")
d1.plot(figsize=(10, 4), title="First difference"); plt.show()

# If still non-stationary: log transform, then difference / second difference
d2 = series.diff().diff().dropna(); adf(d2, "d=2")
logd = np.log(series).diff().dropna(); adf(logd, "log + d=1")
d = 1 if adfuller(d1)[1] < 0.05 else 2   # chosen differencing order

# Train / test split on the original series
```

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n = int(len(series) * 0.8)
train, test = series[:n], series[n:]

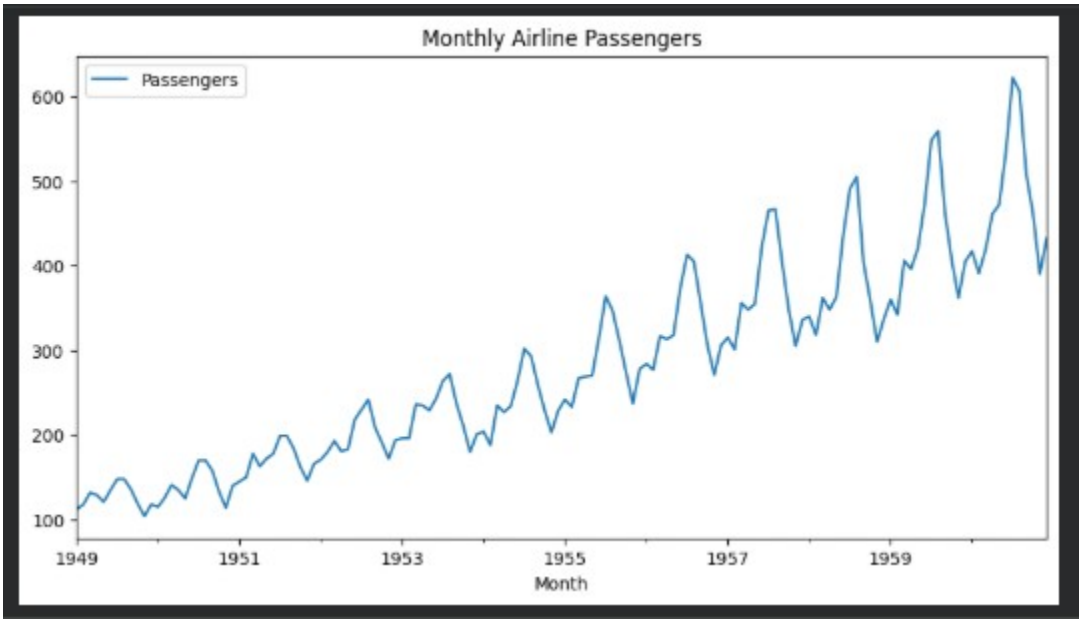
model = ARIMA(train, order=(2, d, 2)).fit()
print(model.summary())
pred = model.forecast(steps=len(test))

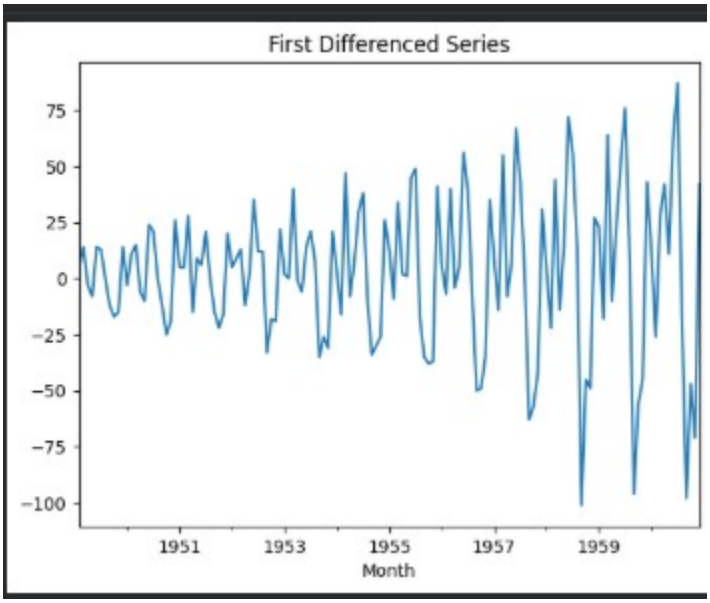
mse = mean_squared_error(test, pred)
print("MSE :", mse)
print("MAE :", mean_absolute_error(test, pred))
print("RMSE:", np.sqrt(mse))

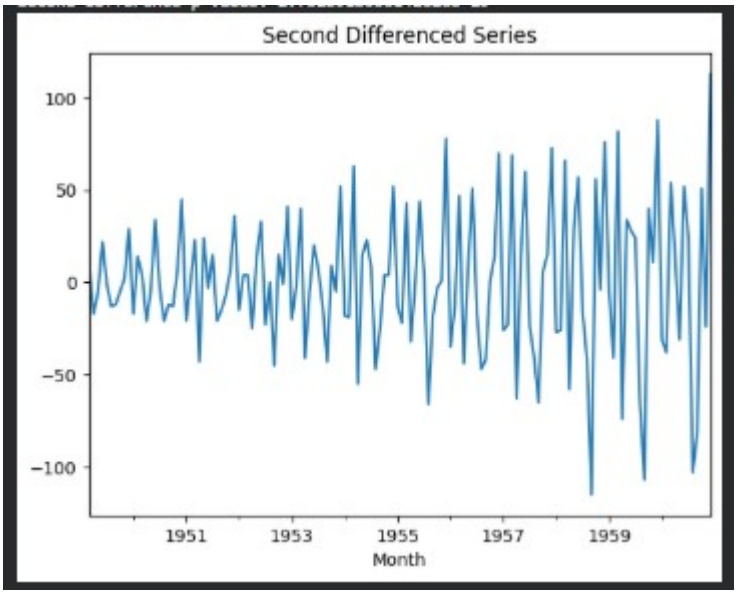
plt.figure(figsize=(10, 5))
plt.plot(train, label="Train"); plt.plot(test, label="Test"); plt.plot(pred,
label="Forecast")
plt.legend(); plt.title("ARIMA Forecast"); plt.show()
```

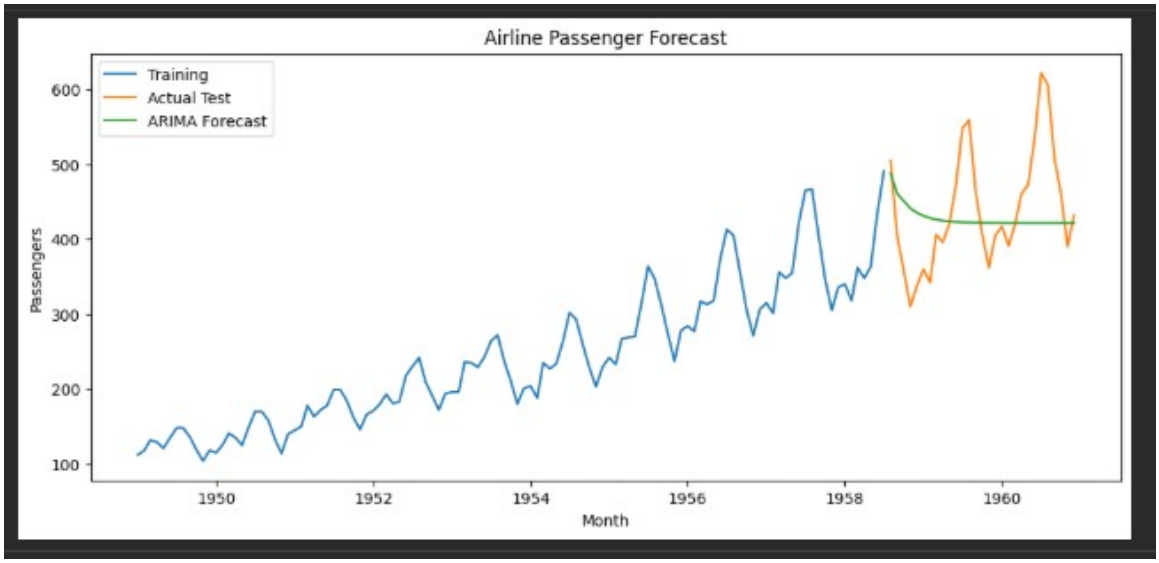
Result

The ADF test showed the original series to be non-stationary; differencing made it stationary, and the ARIMA model fitted on the training data produced forecasts for the test period. MSE, MAE and RMSE were computed, and the train, test and forecast curves were plotted together.




First Differenced Series
75
50
25
0
−25
−50
−75
−100
1951
1953
1955
1957
1959
Month


Second Differenced Series
100
50
0
−50
−100
1951
1953
1955
1957
1959
Month


Airline Passenger Forecast
Training
Actual Test
ARIMA Forecast
Passengers
Month
100
200
300
400
500
600
1950
1952
1954
1956
1958
1960